PREDICTING RECIPE DIFFICULTY FROM TEXT USING MACHINE LEARNING

Objective Difficulty Ratings Through NLP And Classification

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Problem

- Inconsistent criteria across platforms
- Labels vary (AllRecipes vs. Food.com)
- Users struggle to match recipes to skill level
- No objective standard for difficulty

Goal: Create an objective, data-driven system that predicts difficulty from recipe content alone.

Dataset

- RecipeNLG: 30,000 recipes from 2.2M+ corpus
- Split: 70% train / 15% validation / 15% test
- Features: Title, ingredients, directions, source

Labeling Strategy

EASY (22%):

- ≤5 steps, ≤8 ingredients
- ≤15 avg words per step
- No advanced techniques

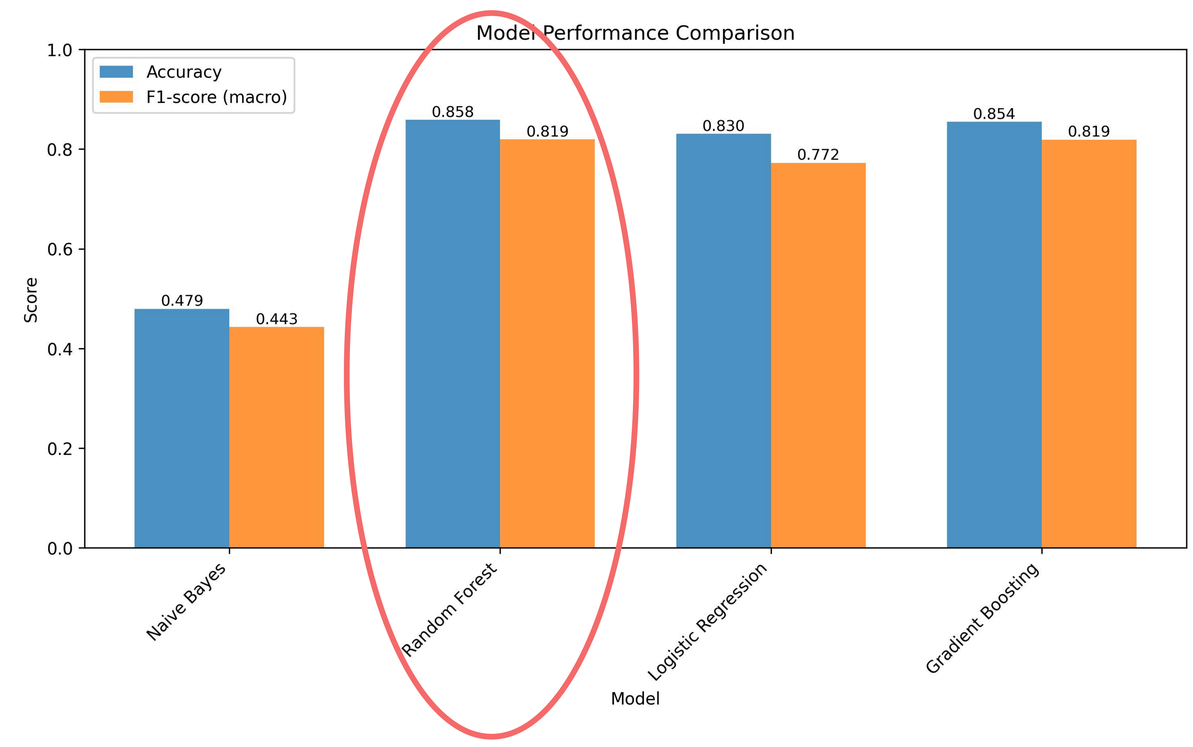
HARD (9%):

- Steps > 12 AND words/step > 18 AND cooking_verbs > 3
- OR ingredients > 17
- OR advanced techniques

MEDIUM (69%):

- Everything else

Recipe Difficulty Can Be Predicted From Text Alone With 86% Accuracy



Features & Models

Text Features:

- 1000 TF-IDF features (unigrams + bigrams)
- Automatically selects informative terms

Numeric Features:

- Text length, cooking verb count
- Excluded labeling features (no data leakage)

Models:

- Random Forest (100 trees)
- Gradient Boosting (100 estimators)
- Logistic Regression
- Naive Bayes (baseline)

Results

Test Set Performance:

- Random Forest: 85.8% accuracy, 0.82 F1
- Gradient Boosting: 85.4% accuracy, 0.82 F1
- Logistic Regression: 83.0% accuracy, 0.77 F1

Per-Class Performance (RF):

- Easy: 73% F1 (77% precision, 69% recall)
- Medium: 90% F1 (87% precision, 93% recall)
- Hard: 83% F1 (96% precision, 72% recall)

Conclusion & Future Work

Key Findings:

- 86% accuracy from text alone
- Models learn semantic patterns, not rules
- Balanced performance (F1: 73–90%)
- Ready for recipe platforms

Future Directions:

- User feedback integration
- Advanced NLP (BERT, RoBERTa)
- Multi-modal features (images, prep time)
- Domain-specific models

Data Sources & References

Dataset:
RecipeNLG Dataset (https://recipenlg.cs.put.poznan.pl/)

Related Work:
Kazuma Kusu, Nozomi Makino, Takamitsu Shioi, and Kenji Hatano. 2017. Calculating Cooking Recipe's Difficulty based on Cooking Activities. In Proceedings of the 9th Workshop on Multimedia for Cooking and Eating Activities in conjunction with The 2017 International Joint Conference on Artificial Intelligence (CEA2017). Association for Computing Machinery, New York, NY, USA, 19–24. https://doi.org/10.1145/3106668.3106673
Chen, J. (n.d.). Predicting Cuisine Regionality Based on Recipe Ingredients Using NLP. Medium. https://medium.com/@jaimejcheng/predicting-cuisine-regionality-based-on-recipe-ingredients-using-nlp-5262e83884eb